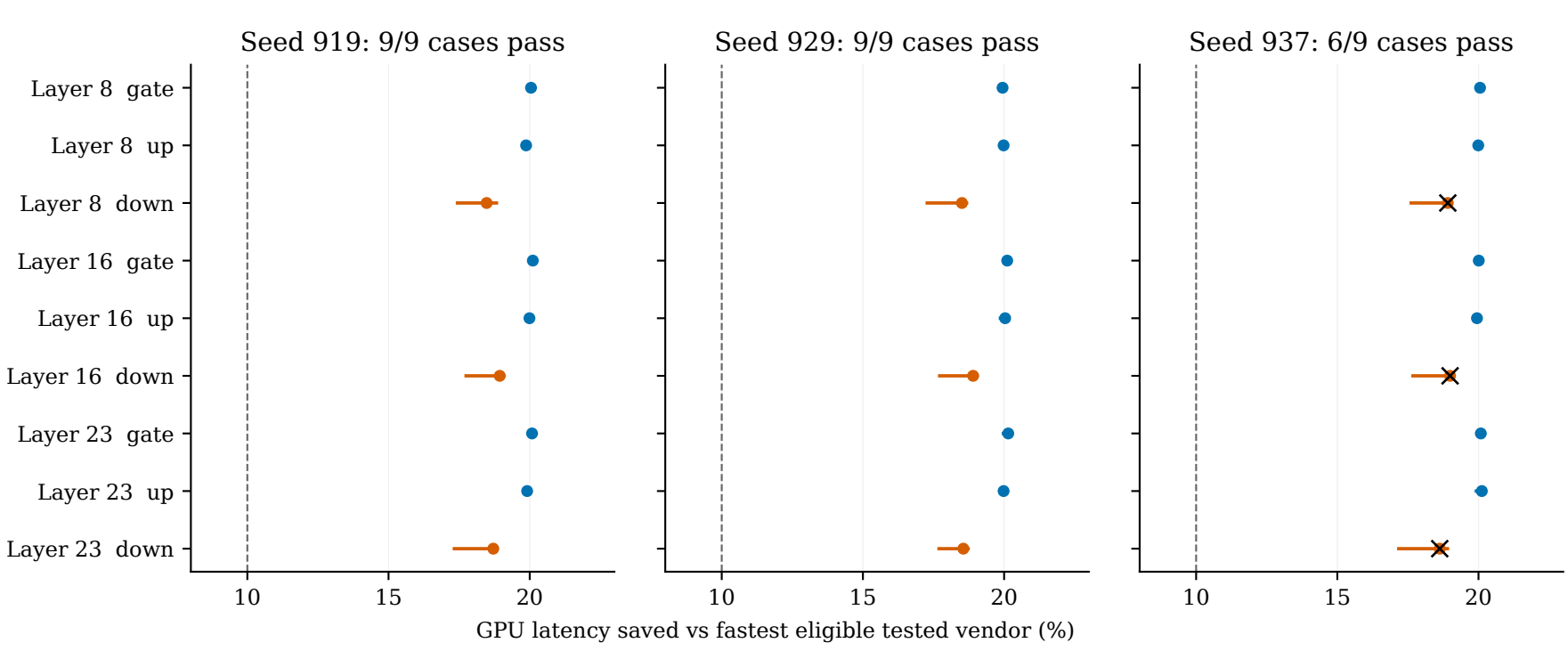
Held-out large M=1 projections
Native BF16 input/weights · FP32 accumulation · FP32 output · 31 paired rounds/case
Seed 919: 9/9 cases pass
Seed 929: 9/9 cases pass
Seed 937: 6/9 cases pass
Layer 8 gate
Layer 8 up
Layer 8 down
Layer 16 gate
Layer 16 up
Layer 16 down
Layer 23 gate
Layer 23 up
Layer 23 down
10
15
20
GPU latency saved vs fastest eligible tested vendor (%)
Bars: 95% paired bootstrap intervals. Dashed line: >10% target. Black x: failed kernel gate.
Performance profile · AC online · stock · NVIDIA GeForce RTX 3070 Ti Laptop GPU · memory 810 MHz